



Abhishek Kumar

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Profile Summary

Final-year B.Tech student at NIT Calicut with a strong command of Data Structures and Algorithms, Object-Oriented Programming, and systems programming in C++. Built projects in network security (Deep Packet Inspection Engine) and machine learning (Neural Network Spam Classifier), applying core CS concepts including Operating Systems and Computer Networks. Active LeetCode practitioner; looking to join a high-growth product company as a Software Development Engineer and contribute to large-scale, efficient backend systems.

Education

National Institute of Technology, Calicut (NIT Calicut)	2023-2027
B.Tech in Electrical and Electronics Engineering CGPA: 7.22/10	
Krishna Public School, Patna	2023
Class 12- CBSE Percentage: 90.4%	
SKP Vidya Vihar, Deoghar	2021
Class 10- CBSE Percentage: 91.3%	

Technical Skills

Programming Languages: C++, Python, JavaScript

Data Structures & Algorithms: Arrays, Linked Lists, Binary Trees, Stacks, Queues, Graphs, Dynamic Programming, Hashing, Sorting & Searching

Core CS Fundamentals: Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, TCP/IP

Web Technologies: HTML, CSS, JavaScript

Tools & Platforms: VS Code, MATLAB, LT Spice, Simulink, Git

Libraries & Frameworks: Scikit-learn, NumPy, Pandas

Projects

Deep Packet Inspection (DPI) Engine | C++, PCAP, Computer Networks :-Developed a C++-based network traffic analyzer for offline packet inspection using PCAP files. Parsed Ethernet, IPv4, TCP, UDP, TLS, and QUIC packets to extract protocol metadata, source/destination information, and connection details. Implemented checksum validation, SNI extraction, and a modular packet-processing pipeline for traffic analysis and protocol inspection. Applied concepts from Computer Networks, Deep Packet Inspection, and Cybersecurity to classify network traffic and analyze encrypted and unencrypted communications.

Spam Email Classifier | Python, Scikit-learn, NumPy, Pandas, OOP 2024 :-Designed and implemented a spam email detection system using a backpropagation-based neural network, applying Object-Oriented Programming principles to structure model components into reusable classes. Built end-to-end data pipeline including preprocessing, TF-IDF feature extraction, and model training; tuned hyperparameters to optimize classification accuracy on real-world email datasets. Analyzed model performance using precision, recall, and F1-score metrics; applied knowledge of supervised learning and statistical evaluation to improve generalization on unseen data.

Positions Of Responsibility

Operations & Logistics Lead | RAGAM Cultural Festival, NIT Calicut

Managed end-to-end logistics and crowd coordination for events with 1,000+ attendees, demonstrating ownership and execution under pressure.

Event Coordinator | RAGAM, Durga Puja, Holi — NIT Calicut

Led diverse teams across multiple campus cultural events, ensuring timely execution and strong student engagement.

Creative Promotion Lead | Tathva Technical Fest, NIT Calicut

Designed and edited promotional content for the annual tech fest, growing social media reach on Instagram and WhatsApp.